

SARAL SINGHAL

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Education

Vellore Institute of Technology

Bachelor's in Computer Science CGPA: 8.4/10

Bhopal, Madhya Pradesh

2023 – 2027

Technical Skills

Languages: Java, Python, SQL

ML Frameworks: TensorFlow, Scikit-Learn, Keras, PyTorch

Data and Analytics: Pandas, NumPy, SHAP, Matplotlib

Concepts: Data Cleaning, EDA, Feature Engineering, Ensemble Learning

Tools & Software: Git, GitHub, Jupyter Notebook, VS Code, Oracle BI, N8N, Power BI

Experience

Deloitte

Intern - Strategy Risk Transactions

Gurugram, India

May 2026 – Present

- **Report Building:** Developed research and analytical studies on Public-Private Partnership (PPP) models and data centre infrastructure development in India.
- **Asset Review System:** Built a Power BI dashboard for the Asset Registration System under the Asset Monetization Pipeline, enabling seamless and efficient review of resources.

Ernst & Young

Research Intern

Delhi, India

May 2025 – June 2025

- **Market Analysis:** Performed a detailed assessment of non-fare box revenue streams for Chennai Metro Rail Limited (CMRL), evaluating commercial space utilization and land value capture mechanisms.
- **System Specification:** Designed functional requirements and workflow models for a software platform to automate vendor-based non-fare revenue collection and monitoring.
- **Technical Documentation:** Mapped Chhattisgarh and Uttar Pradesh National Highways through GIS mapping for Asset Registration.

Projects

BleedDetectX Python, TensorFlow, Keras, PyTorch

[Link](#)

- **Medical Imaging Models:** Developed deep learning architectures including U-Net, ResNet-UNet, and DMP-WCEBleedNet for bleeding classification and segmentation in Wireless Capsule Endoscopy images.
- **High-Performance Results:** Achieved peak performance using DMP-WCEBleedNet with a Dice Coefficient of 0.812 and classification accuracy of 98.31% on the test dataset.
- **Multi-Task Learning:** Built a PyTorch-based ResNet-UNet leveraging a pre-trained ResNet34 encoder for simultaneous bleeding classification and segmentation.
- **Comprehensive Evaluation:** Deployed DMP-WCEBleedNet with multi-metric assessment including Dice, IoU, Precision, Recall, and F1-Score to ensure clinical-grade reliability.

ChurnSense Python, XGBoost, SMOTE, SHAP

[Link](#)

- **Customer Retention:** Proposed a decision intelligence platform for customer retention that combines churn-risk assessment, customer lifetime value analysis, ROI optimization, and strategic customer segmentation to support targeted business interventions.
- **Machine Learning:** Implemented an ensemble churn prediction pipeline using Logistic Regression, Random Forest, and XGBoost with SMOTE balancing and SHAP explainability, deployed through an interactive Streamlit dashboard for analytics and decision support.

Certifications

- Microsoft Certified: Data Engineer Associate
- Oracle Analytics Cloud Certified Professional
- Google Skill Badge: Inspect Rich Documents with Gemini Multimodality and Multimodal RAG (Vertex AI)
- Google Skill Badge: Prompt Design in Vertex AI
- Applied Machine Learning Specialization – Coursera